

# **Development of a Library Book Borrowing System**

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## **I. Introduction**

Technology continues to advance rapidly, leading to the development of modern web applications that improve efficiency, accuracy, and user experience. Libraries play an important role in educational institutions by providing access to learning resources. However, manual management of book borrowing and returning can be time-consuming and prone to errors. This project proposes the development of a Library Book Borrowing System, a web-based application designed to streamline library operations. The system will enable librarians and users to manage book records, monitor borrowing and returning transactions, and track due dates efficiently. Through this project, modern web development technologies will be utilized to create a functional, user-friendly, and reliable system within the allotted development period.

## **II. Objectives**

1. To develop a Library Book Borrowing System for efficient library management.
2. To apply the fundamental concepts of application development.
3. To automate the borrowing and returning processes of library books.
4. To maintain accurate records of book availability and transactions.
5. To complete a functional and user-friendly system within one week.

## **III. Scope of the System**

- Users can add, edit, and delete book records.
- Users can borrow and return books.
- The system automatically tracks due dates.
- Data is stored using a MySQL database.
- The system is accessible through a web browser.

## **IV. Limitations**

- No user account or login system.
- No barcode or RFID scanning.
- No email or SMS notifications for due dates.
- Limited design and advanced features due to time constraints.

## **V. Features of the System**

Add books, edit book information, delete books, borrow books, return books, due date tracking, display borrowed books, show due dates, and highlight overdue books.

## **VI. System Process Flow**

1. User adds a book to the system.
2. System saves the book in the database.
3. User selects a book to borrow.
4. System records borrowing details and due date.
5. System updates the book status as available or borrowed.
6. User returns the book.
7. System updates the record and marks it as available.

## **VII. Tools and Technologies**

HTML and CSS for the user interface, JavaScript and PHP for system functionality, MySQL for database management, and Visual Studio Code for development.

## **VIII. Conclusion**

This project demonstrates how a system can improve library operations through the use of modern web development technologies. By developing a Library Book Borrowing System, users will be able to manage books more efficiently through organized and automated processes. The project will help enhance programming skills and deepen understanding of web development and database technologies while delivering a practical and user-friendly solution.